under a hierarchical Bayesian model using Markov Chain Monte Carlo, assuming partial pooling of parameter estimates across participants (see Supplemental Materials).

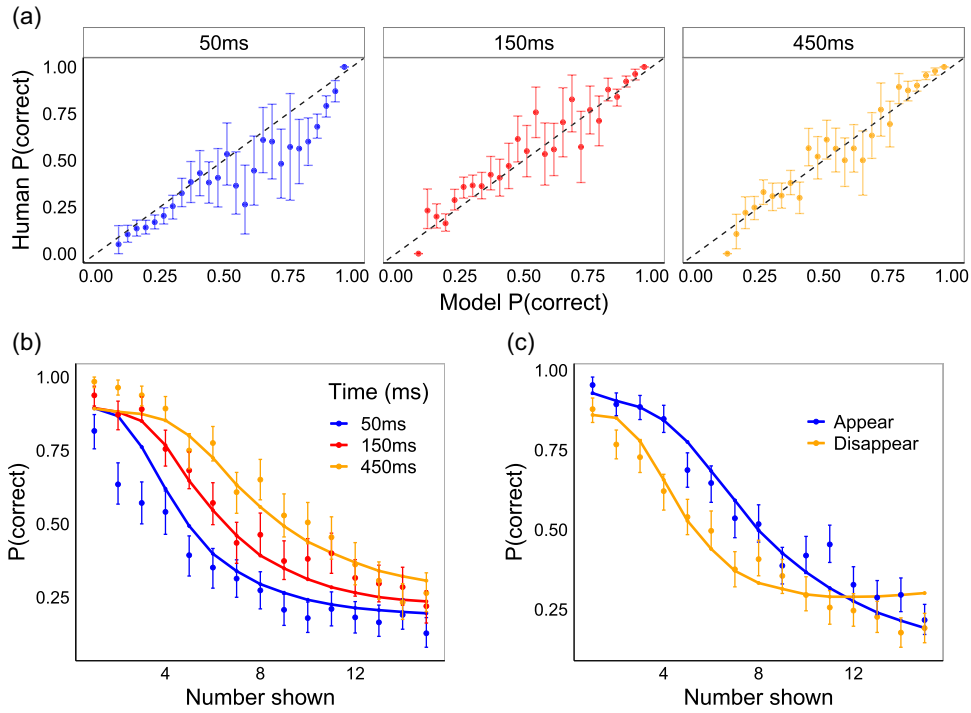
The maximum a posteriori (MAP) estimates for the group-level parameters were $a = 33.5$ (CI [32.2, 34.6]), $k = 0.21$ (CI [0.20, 0.22]), $p_g = 0.16$ (CI [0.12, 0.19]), and $\alpha = .35$ (CI [0.33, 0.37]). This entails information bounds of 17.9, 22.5, and 28.3 bits at 50 ms, 150 ms, and 450 ms, respectively. The relatively high inferred rate of guessing likely reflects the fact that the model does not account for spatial errors, treating each cell independently. Figure 5a shows the model's predicted accuracy (x-axis) against human performance (y-axis) across all exposure durations (facets). Comparing the points to the dashed $y = x$ line reveals that the model's predictions tightly align with human accuracy across exposure durations, though the model is slightly biased to overestimate human performance at short times (left facet). The correlation between model predictions and human data across trials grouped by numerosity and exposure duration was 0.96 ($R^2 = 0.93$), another indication that the model provides a good fit to the data.

In line with participants, the model predicts near-veridical memory for visual displays with small numbers of objects, at longer exposure durations, and sharply increasing noise for larger numbers of objects and shorter durations. Figure 5b shows human accuracy (points and error bars), the model's predicted accuracy (lines) as a function of the total number of cells filled in, grouped by the exposure duration (colors). As predicted by the model, participants' performance saturates only for small numerosities at longer durations and quickly degrades as a function of number in each duration. The one notable discrepancy is that the model predicted better performance on small numerosities ($n < 4$) at 50 ms than was actually observed. Figure 5c depicts accuracy grouped by whether a cell appeared or disappeared from the first to second display and shows that participants performed substantially better on "appear" trials than "disappear" trials—a trend the model captures. The model would capture this trend even if α was fixed to 0.5, and in fact, higher values of α exaggerate rather than reduce the gap between "appear" and "disappear" trials.

To be clear, the fact that human performance on the change-localization task is strongly affected by numerosity is not an indication that the visual system is representing or using number. In fact, the experiment was explicitly designed so that number could not be used as a heuristic: Participants always knew that there was either one more gray cell on the second display than the first display (on "appear" trials) or one fewer (on "disappear" trials). Instead, the effect of numerosity on performance is an indication that spatial memory is making use of limited information in an efficient way,

Figure 5

Model Predictions and Data From Localization Task (Experiment 1)



Note. (a) Binned (25 bins/facet) model predictions (x-axis) and human data (y-axis) of performance on the change-localization task. Each facet shows predictions at different exposure durations. In (b) and (c) model, predictions are shown as lines, and human data from the change-localization task are shown as points with bootstrapped 95% confidence intervals. (b) Accuracy (y-axis) in the change-localization task as a function of the number of grid cells filled (x-axis) at each exposure duration. (c) Accuracy (y-axis) as a function of number (x-axis) grouped by whether or not a cell appeared or disappeared from first-to-second presentation. See the online article for the color version of this figure.

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